

CELINE NGUYEN

9093 Heron Avenue, Fountain Valley, CA 92708

📞 714-726-9113 ✉️ can019@ucsd.edu [in linkedin.com/in/celineenguyen/](https://www.linkedin.com/in/celineenguyen/) github.com/celineenguyen

Education

University of California San Diego

Bachelor of Science in Mathematics-Computer Science

June 2027

La Jolla, California

Relevant Coursework

- Data Structures
- Object Oriented Design
- Graph Theory
- Algorithms
- Probability / Statistics
- Linear Optimization
- Mathematical Reasoning

Projects

Nutribase | React, Node.js, Express, WebSockets, ESP32

April 2025 – June 2025

- Designed and implemented a real-time React dashboard visualizing weight trend data using Chart.js.
- Integrated a Node.js/Express backend with WebSocket streams to process and display live sensor data from ESP32.
- Architected full-stack data flow from embedded device to frontend, ensuring low-latency updates.
- Built reusable Material-UI components to improve scalability and maintainability of the interface.

Diamond Duck | React, Tailwind CSS

April 2025

- Developed responsive and accessible UI components using React and Tailwind CSS to ensure cross-device compatibility.
- Designed and implemented routing and navigation architecture to improve usability and information flow.
- Structured reusable component hierarchy to support scalable feature expansion.

Research Experience

Sleep Duration and Academic Performance Study | COGS 108, UCSD

January 2026 – Present

- Formulated research question examining statistical differences in CGPA between students sleeping 7+ hours and those sleeping fewer than 7 hours.
- Performed exploratory data analysis on 600+ student observations from the CMU Sleep Dataset (Kaggle), including data cleaning and variable operationalization.
- Implemented hypothesis testing framework to evaluate the significance of CGPA differences across sleep-duration groups.
- Assessed potential confounders including gender, first-generation status, and underrepresented group indicators.

Technical Skills

Languages: Python, Java, C++, C, MATLAB, HTML/CSS

Frameworks & Libraries: React, Node.js, Express, NumPy, pandas, scikit-learn, Matplotlib, Tailwind CSS

Developer Tools: Git, GitHub, Linux, VS Code, Jupyter Notebook, LaTeX

Data & Analysis: Excel (Regression, Solver)

Leadership / Extracurricular

California Scholarship Federation

September 2023 – June 2024

President

- Managed structured attendance tracking system to maintain accurate member records.
- Led development and maintenance of the organization website to increase engagement.
- Coordinated college field trips and organized logistics for student participation.
- Directed fundraising initiatives to support scholarship awards.

Little Chopin Piano School

September 2021 – June 2024

Theory Assistant/Tutor

- Organized and coordinated piano recitals, ensuring smooth event execution.
- Hosted recital events and provided performance support to students.
- Mentored lower-level students and provided structured feedback to improve theory comprehension.
- Graded theory assessments and delivered targeted feedback to enhance performance.

Huntington Beach Public Library

September 2022 – June 2024

Teen Tech Volunteer

- Assisted patrons with technology-related inquiries and digital resource navigation.
- Maintained and troubleshooted library computers, printers, and technical equipment.
- Provided hands-on support to ensure efficient and accessible use of library technology.